

OpenIntro Applied Statistics

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Table of contents

Preface	1
1 Introduction to data	3
1.1 Data basics	3
1.1.1 Observations, variables, and data matrices	3
1.2 Scatterplots and Associations	4
1.3 Exercises	4
2 Summary & Financial Case Study	5
2.1 Quarterly Performance Analysis	5
References	7

Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

[1] 2

1 Introduction to data

Scientists seek to answer questions using rigorous methods and careful observations Chapter 1. Data collected from field notes, surveys, and experiments form the backbone of statistical investigations.

1.1 Data basics

Effective organization and description of data is a first step in most analyses.

1.1.1 Observations, variables, and data matrices

A **data matrix** organizes cases as rows and characteristics as columns.

Table 1.1: Sample records from a peer-to-peer lending portfolio

loan_amount	interest_rate	term	grade	homeownership
22000	10.90	60	B	rent
6000	9.92	36	B	rent
25000	26.30	36	E	mortgage
15000	6.08	36	A	mortgage

As illustrated in Table 1.1, each row corresponds to an observational unit.

DATA MATRIX

Each row of a data matrix corresponds to a unique case (observational unit), and each column corresponds to a variable.

EXAMPLE 1.1: CLASSIFYING VARIABLES

Example problem: In the loan sample shown above, classify `loan_amount` and `homeownership` as either numerical or categorical.

Solution to the example: `loan_amount` takes numerical dollar values where arithmetic is sensible, so it is numerical. `homeownership` categorizes borrowers into distinct groups (rent or mortgage), making it categorical.

GUIDED PRACTICE 1.2: PROPORTION OF RENTERS

What proportion of the loans shown in Table 1.1 represent borrowers who rent their homes?¹

¹Two out of four loans in the sample represent renters: $2/4 = 0.50$ or 50%.

1.2 Scatterplots and Associations

Scatterplots provide a case-by-case view of data for two numerical variables.

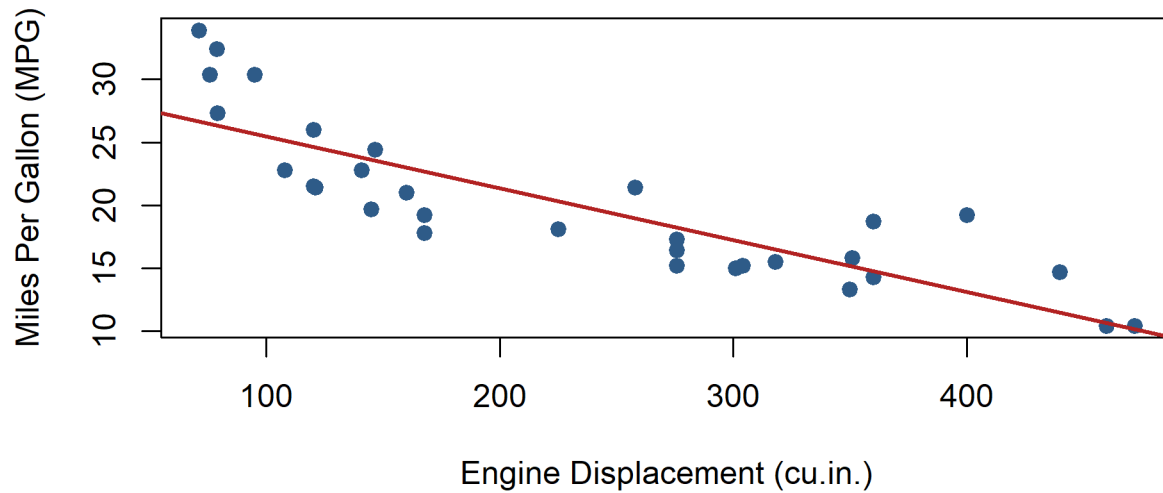


Figure 1.1: Engine displacement versus fuel economy for passenger vehicles

As demonstrated in Figure 1.1, there is a clear downward trend: vehicles with larger engine displacement yield lower fuel economy.

1.3 Exercises

1.1 Sampling methods. A survey selects 100 students at random from a university directory. (a) Identify the target population and the sample. (b) Is this an observational study or an experiment?

2 Summary & Financial Case Study

This chapter illustrates loading external CSV datasets and formatting publication-ready accounting tables.

2.1 Quarterly Performance Analysis

We load company financials from an external CSV file stored in the project data directory.

```
# Read the external CSV file
fin_data <- read.csv("data/financials.csv")

# Compute Net Profit dynamically
fin_data$Net_Profit <- fin_data$Revenue -
  fin_data$Expenses

# Display with knitr::kable
knitr::kable(fin_data)
```

Table 2.1: Quarterly Revenue vs Expenses (USD)

Quarter	Revenue	Expenses	Net_Profit
Q1	45000	31000	14000
Q2	52000	34000	18000
Q3	61000	39000	22000
Q4	78000	42000	36000

As highlighted in Table 2.1, profitability expanded across all four operating quarters.

References

